

CXL 331	WEB PROGRAMMING LAB	CATEGORY	L	T	P	CREDIT	YEAR OF INTRODUCTION
		PCC	0	0	3	2	2021

Preamble: With a dynamic learn-by-doing focus, this laboratory course encourages the students to explore the designing of web application by implementing the relevant and recent techniques. This course challenges the students to exercise their creativity in both programming and designing.

Prerequisite: Basic understanding of computer programming, Internet and Database.

CO1	Explore and develop interactive web pages using markup languages. (Cognitive Knowledge Level: Apply)
CO2	Implement client-side validation using scripting languages using Javascript and PHP (Cognitive Knowledge Level: Apply)
CO3	Design and develop web pages and implement databases connectivity using PHP and MySQL (Cognitive Knowledge Level: Apply)
CO4	Develop and execute Client-side & Server-side scripts using CSS, Javascript and PHP. (Cognitive Knowledge Level: Apply)
CO5	Develop Web Applications using Laravel framework (Cognitive Knowledge Level: Apply)

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

Abstract POs defined by National Board of Accreditation			
PO#	Broad PO	PO#	Broad PO
PO1	Engineering Knowledge	PO7	Environment and Sustainability
PO2	Problem Analysis	PO8	Ethics
PO3	Design/Development of solutions	PO9	Individual and team work
PO4	Conduct investigations of complex problems	PO10	Communication
PO5	Modern tool usage	PO11	Project Management and Finance
PO6	The Engineer and Society	PO12	Lifelong learning

Assessment Pattern

Bloom's Category	Continuous Assessment Test (Internal Exam) <i>Percentage</i>	End Semester Examination <i>Percentage</i>
Remember	20	20
Understand	20	20
Apply	60	60
Analyse		
Evaluate		
Create		

Mark Distribution

Total Marks	CIE Marks	ESE Marks	ESE Duration
150	75	75	3 hours

Continuous Internal Evaluation Pattern:

Attendance : 15 marks
 Continuous Evaluation in Lab : 30 marks
 Continuous Assessment Test : 15 marks
 Viva-voce : 15 marks

Internal Examination Pattern: The marks will be distributed as Algorithm 30 marks, Program 20 marks, Output 20 marks and Viva 30 marks. Total 100 marks which will be converted out of 15 while calculating Internal Evaluation marks.

End Semester Examination Pattern: The marks will be distributed as Algorithm 30 marks, Program 20 marks, Output 20 marks and Viva 30 marks. Total 100 marks will be converted out of 75 for End Semester Examination.

Fair Lab Record:

All Students attending the Web Programming Lab should have a Fair Record. The fair record should be produced in the University Lab Examination. Every experiment conducted in the lab should be noted in the fair record. For every experiment in the fair record the right-hand page should contain Experiment Heading, Experiment Number, Date of Experiment, Aim of Experiment, Details of Experiment including algorithm and Result of Experiment. The left-hand page should contain a print out of the code used for the experiment and sample output obtained for a set of input.

Syllabus

1. Introduction To Web: Client/Server concepts, Components of Web Application, Types of Web Content, Overview of HTTP - HTTP request – response, Generation of dynamic Web Pages, Application Servers, Web Security.
2. Markup Language (HTML): Formatting and Fonts, Commenting Code, Anchors, Backgrounds, Images, Hyperlinks, Lists, Tables, Frames, HTML Forms.
3. Cascading Style Sheet (CSS): The need for CSS, Basic syntax and structure, Inline Styles, Embedding Style Sheets, Linking External Style Sheets, Backgrounds, Manipulating text, Margins and Padding, Positioning using CSS.
4. Client-Side Scripting using JavaScript: Core features, Data types and Variables, Operators- Expressions and Statements, Functions, Objects, Array, String - Date and Math related Objects, Document Object Model, Event Handling, Form handling and validations.
5. An overview of Relational Database Design: Tables, Attributes, Tuples, Primary keys, Foreign keys, Indexes, DDL Commands – CREATE, ALTER, DROP and TRUNCATE; DML Commands – SELECT, INSERT, UPDATE and DELETE.

6. Server-Side Scripting using PHP: Setting up the environment (Example - XAMP server), PHP Programming basics - Print/echo, Variables and constants, Strings and Arrays, Operators, Control structures and looping structures, Functions, Reading Data in Web Pages, Embedding PHP within HTML, Establishing connectivity with database, Debugging with phpdbg.
7. Web Application development using Laravel, Naming convention, MVC model, Connectivity with Database, Database interaction.
8. Debugging web apps: Browser debugging tools (Any browser web developer tools) –View and change the DOM and CSS, Console, Debug JavaScript, View and debug network activity, Performance tools etc.

List of Lab Experiments/Exercises

1. Create a simple HTML file to demonstrate the use of different tags. *
2. Create a HTML file to link to different HTML page which contains images, tables and also link within a page. *
3. Create a HTML page with different types of frames such as floating frame, navigation frame & mixed frame.
4. Create a HTML file by applying the different styles using inline, external & internal style sheets.
5. Create a registration form using HTML.*
6. Create a HTML page to explain the use of various predefined functions in a string and math object in java script.*
7. Generate the calendar using JavaScript code by getting the year from the user.
8. Create a HTML registration form and to validate the form using JavaScript code.*
9. Create a HTML page to change the background color for every click of a button using JavaScript Event Handling.*
10. Create a HTML page to display a new image and text when the mouse comes over the existing content in the page using JavaScript Event Handling.
11. Create a HTML page to show online exam using JavaScript.*
12. Develop a registration form using PHP and do necessary validations.*
13. Compose Electricity bill from user input based on a given tariff using PHP.*
14. Build a PHP code to store name of students in an array and display it using print_r function. Sort and Display the same using assort & arsort functions.*
15. Build a PHP code to store name of Indian Cricket players in an array and display the same in HTML table.*
16. Develop a PHP program to connect to a database and retrieve data from a table and show the details in a neat format.*
17. Develop Web applications using HTML and PHP and deploy.*
18. Using PHP and MySQL, develop a program to accept book information viz. Accession number, title, authors, edition and publisher from a web page and store the information in a database and to search for a book with the title specified by the user and to display the search results with proper headings.*
19. Develop a web application using Laravel and test the application on an Application Server. * (assign as micro project)

Note: Students can be given a group micro project, so that they learn to work in a team environment. They can also be trained on project management tools.

*All programs are mandatory and should be completed in the lab.

Reference Books

1. David Flanagan, “**JavaScript: The Definitive Guide**”, 6th Edition”, O'Reilly Media
2. Douglas E Comer, “**The Internet Book: Everything You Need to Know About Computer Networking and How the Internet Works**”, 4th Edition, Prentice Hall
3. Harvey Deitel and Abbey Deitel, “**Internet and World Wide Web - How To Program**”, 5th Edition, Pearson Education
4. Abraham Silberschatz, Henry F. Korth, S. Sudarshan, “**Database System Concepts**”, McGraw Hill Education, 6th Edition (2011)
5. Steve Suehring, Tim Converse, and Joyce Park, “**PHP6 and MySQL Bible**”, Wiley India Pvt Ltd (2009)
6. Steven Holzner, “**PHP-The Complete Reference**”, Tata McGraw Hill, 1st Edition (2007)
7. Thomas A Powell, Fritz Schneider, “**JavaScript: The Complete Reference**”, 3rd Edition, Tata McGraw Hill.

